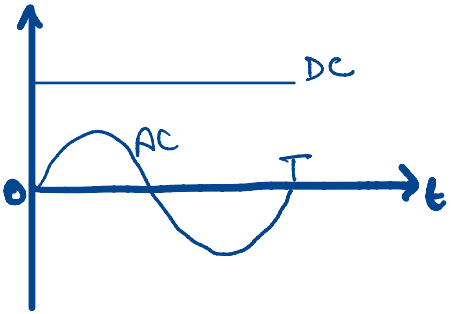
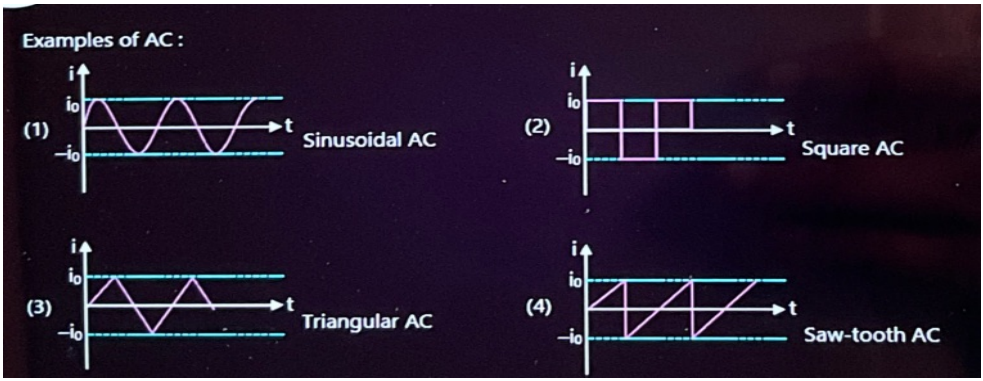
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Alternating Current

An alternating current is that current whose magnitude changes continuously with time and direction reverses periodically.

Direct current is that current, which flows with a constant magnitude in the same direction





When a coil is rotated in a magnetic field an alternating EMF is induced in it which is given by the relation: $\mathcal{E} = E_0 \sin\omega t$

then by ohms law the current in the circuit will be

$$I = \frac{\mathcal{E}}{R} = \frac{E_0}{R}\sin\omega t$$

$$I = I_0 \sin\omega t$$

ω = angular speed or angular frequency

$\boxed{\omega = 2\pi f}$

Thus, the current in the circuit varies sinusoidally with time and is called alternating current

Frequency: the number of cycles completed per second by an alternating current is called its frequency and is denoted by F. The frequency of an alternating current is same as the frequency of rotation of the coil in the magnetic field –

$$f = \frac{1}{T} = \frac{\omega}{2\pi}$$

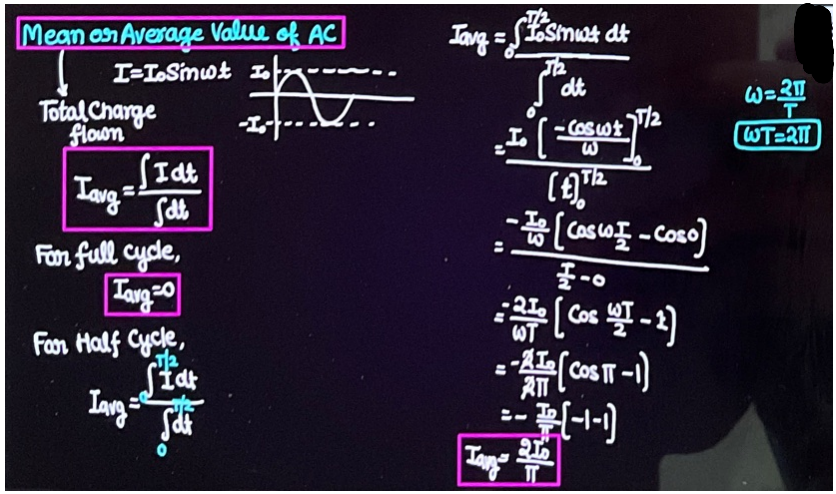
So an alternating current can be represented as

$$I = I_0 \sin\omega t = I_0 \sin 2\pi f t = I_0 \sin\frac{2\pi}{T} t$$

- In India : $f = 50\ Hz$, supply voltage $= 220$ volt
- In USA : $f = 60\ Hz$, supply voltage $= 110$ volt

Mean or average value of A.C.

It is defined as that value of direct current which sends the same charge in a circuit in the same time as is sent by the given alternating current in its half time period. It is denoted by I_m or I_a



$I = I_0 \sin\omega t$

For Half cycle,

$$I_{avg} = \frac{2}{\pi} I_0 = 0.637 I_0 = 63.7\% \text{ of peak current}$$

$V = V_0 \sin\omega t$

For Half cycle,

$$V_{avg} = \frac{2}{\pi} V_0 = 0.637 V_0 = 63.7\% \text{ of peak voltage.}$$

Root mean square (RMS) or virtual or effective value of A.C.

It is defined as that value of a direct current which produces the same heating effect in a given registered as is produced by the given alternating current and past for the same time. It is denoted by I_{rms}, I_v or by I_{eff}.

RMS Value of AC

Heat Generated

$$I_{rms} = \sqrt{\frac{\int I^2 dt}{\int dt}}$$

For full cycle.

$$\int_0^T I^2 dt = \int_0^T (I_0 \sin\omega t)^2 dt$$

$$= I_0^2 \int_0^T \sin^2\omega t \, dt$$

$\cos 2\theta = 1 - 2\sin^2\theta$

$\sin^2\theta = \frac{1-\cos 2\theta}{2}$

$$\int_0^T I^2 dt = I_0^2 \int_0^T \frac{1-\cos 2\omega t}{2} dt$$

$$= \frac{I_0^2}{2}\left[\int_0^T dt - \int_0^T \cos 2\omega t \, dt\right]$$

$$= \frac{I_0^2}{2}\left[(t)_0^T - 0\right]$$

$$= \frac{I_0^2}{2} T$$

$$\int_0^T dt = (t)_0^T = T$$

$$I_{rms} = \sqrt{\frac{\frac{I_0^2}{2} T}{T}}$$

$$I_{rms} = \frac{I_0}{\sqrt{2}} = 0.707 I_0$$

Similarly

$$V_{rms} = \frac{V_0}{\sqrt{2}} = 0.707 V_0$$

The alternating current and voltage is a generally measured and specified in terms of the RMS values. When we say that the household supply is 220 V AC. We mean that its RMS value is 220 V the peak value would be $V_0 = \sqrt{2}\, V_{rms} = \sqrt{2} \times 220 = 311$ V.

-AC ammeter and voltmeter always measure RMS value

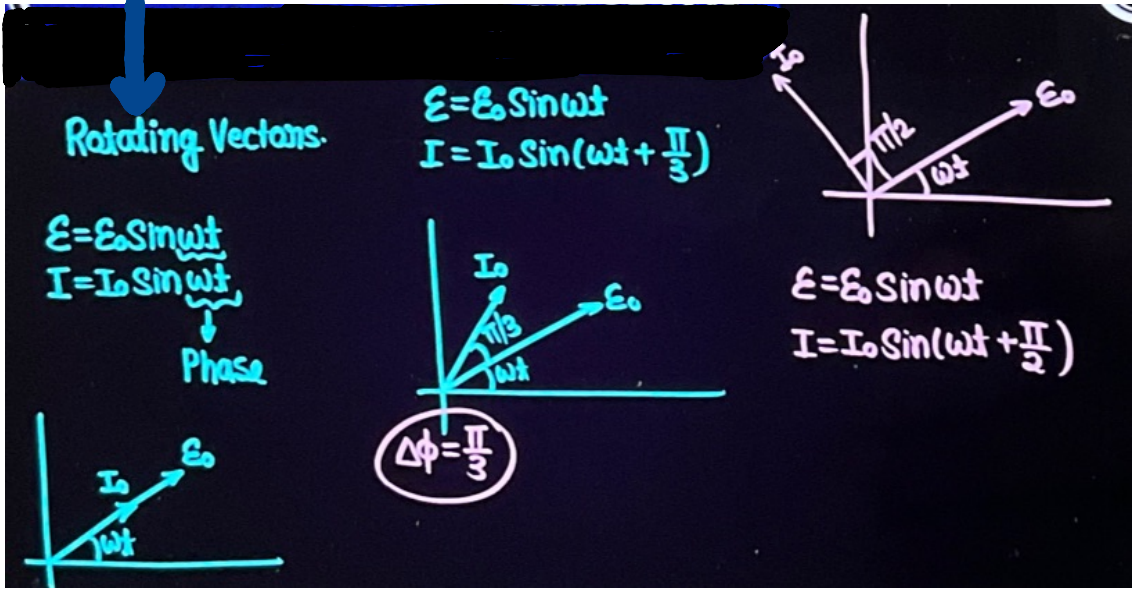
-Values printed on a C circuits are RMS values

$$V_{rms} = 220V$$

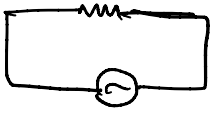
$$\frac{V_0}{\sqrt{2}} = 220V$$

$$\boxed{V_0 = 311V}$$

Phasors and Phasor diagrams



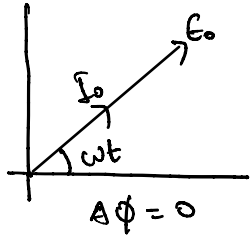
A.C. Circuit containing only resistor



$$\mathcal{E}=\mathcal{E}_0 \sin\omega t$$

$$I=\frac{\mathcal{E}}{R}=\frac{\mathcal{E}_0 \sin\omega t}{R}$$

$$I=I_0 \sin\omega t.$$

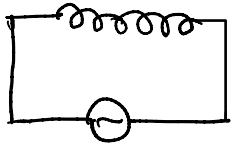


$V=IR$ ohm's law

$|\mathcal{E}|=L\frac{dI}{dt}$ = self induction

$q=CV$

A.C. Circuit containing only Inductor



$E = E_0 \sin\omega t$

$E = L\frac{dI}{dt}$

$E_0 \sin\omega t = L\frac{dI}{dt}$

$\int dI = \frac{E_0}{L}\int \sin\omega t \, dt$

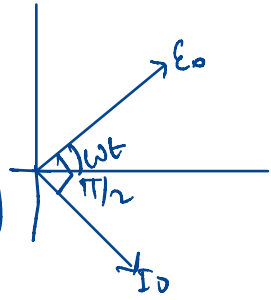
$I = \frac{E_0}{L}\left[-\frac{\cos\omega t}{\omega}\right]$

$-\frac{E_0}{\omega L}\sin\left(\frac{\pi}{2}-\omega t\right)$

$\because \phi = \pi/2$

$I = \frac{E_0}{\omega L}\sin\left(\omega t - \pi/2\right)$

ωL → X_L (Inductive reactance)

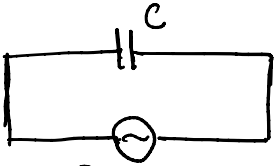


$I = I_0 \sin\left(\omega t - \pi/2\right)$

$\downarrow$

$I_0 = \frac{E_0}{X_L} = \frac{E_0}{\omega L} = \frac{E_0}{2\pi f L}$

A.C. Circuit containing only Capacitor



$E = E_0 \sin\omega t$

$q = CV$

$I = \frac{dq}{dt} = C\frac{d(E_0 \sin\omega t)}{dt}$

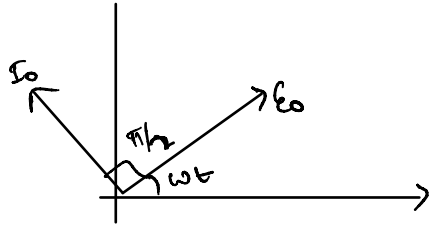
$= E_0 C \cos\omega t \cdot \omega$

$I = \omega C E_0\left[\sin\left(\frac{\pi}{2}+\omega t\right)\right]$

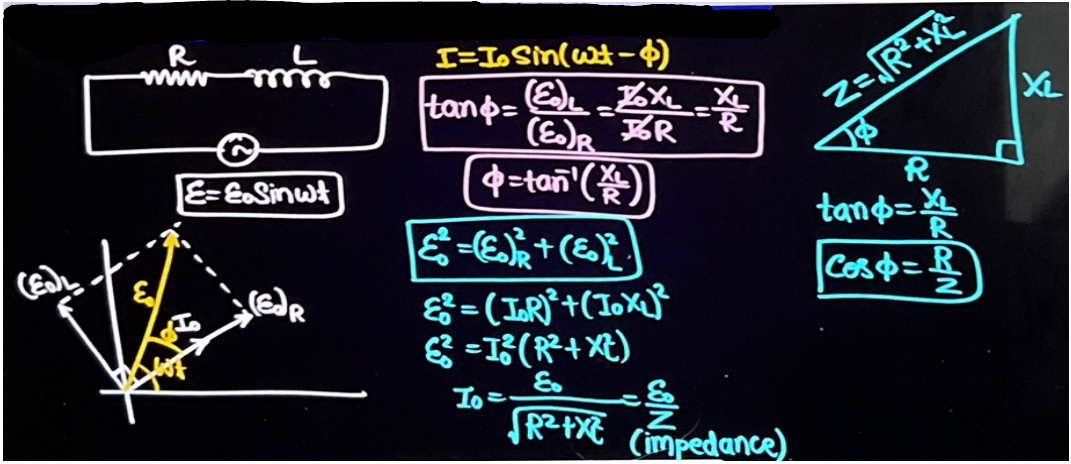
$I = \frac{E_0}{1/\omega C}\sin\left(\omega t + \pi/2\right)$

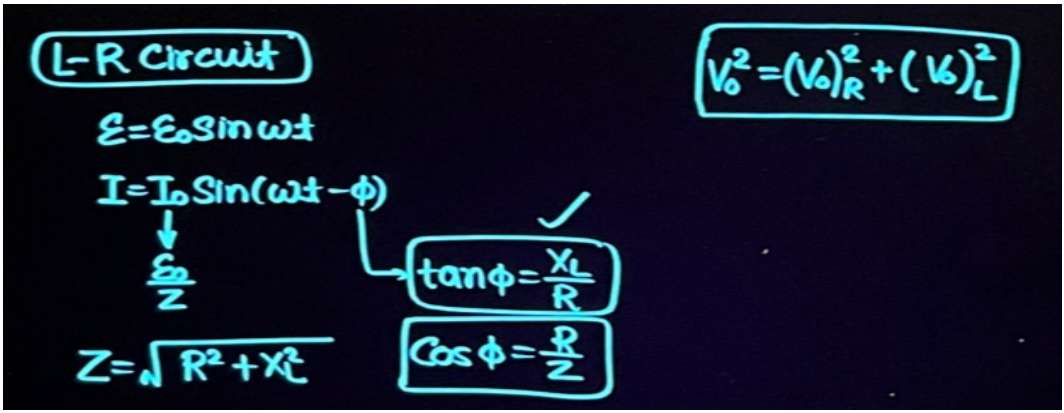
$I = \frac{E_0}{X_C}\sin\left(\omega t + \pi/2\right)$

$I = I_0 \sin\left(\omega t + \pi/2\right)$

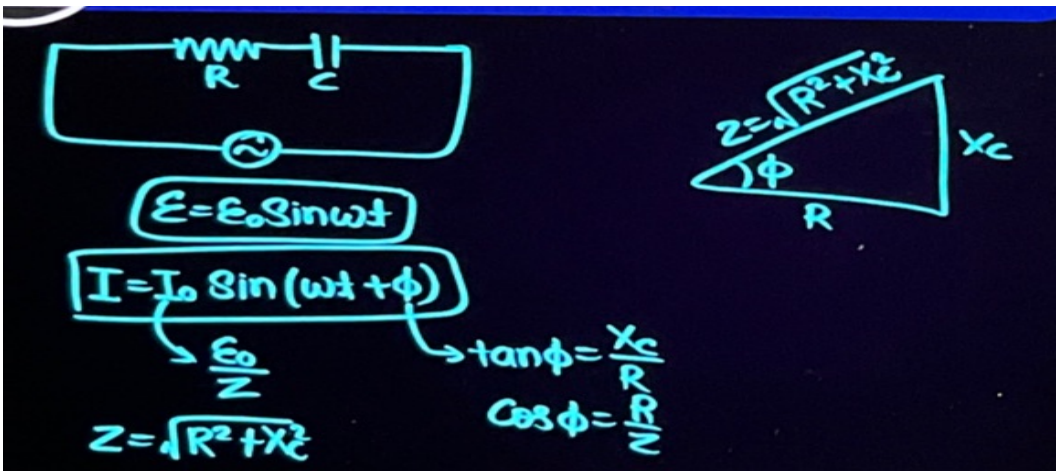


A.C. Circuit with resistance & inductance in series

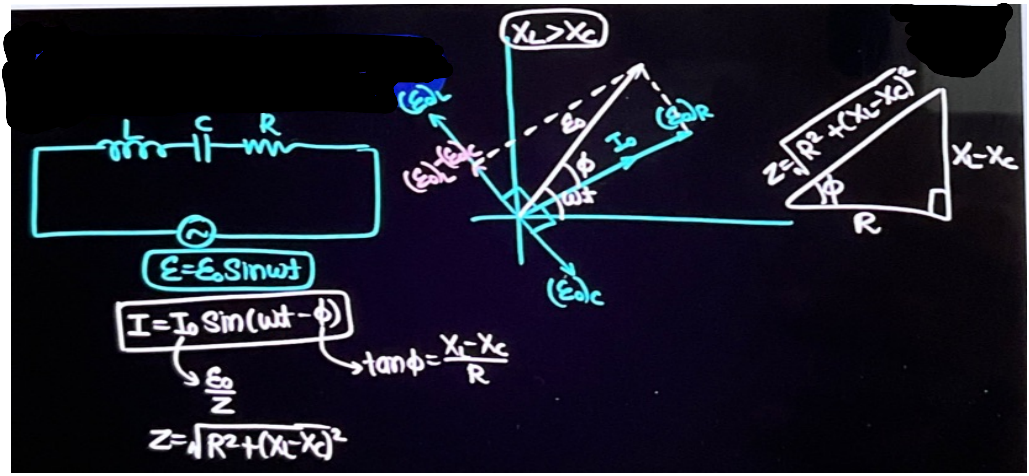




A.C. Circuit with resistance and capacitor in series



Series LCR-circuit



Susceptance: The reciprocal of the reactance of an AC circuit is called its susceptance. Its SI unit is ohm^{-1} or mho

$1/X_L$ or $\frac{1}{X_C}$

Admittance: The reciprocal of the impedance of an AC circuit is called its admittance. Its SI unit is ohm^{-1} or mho

$1/Z$

Resonance condition of series LCR-circuit.

A series LCR circuit is said to be in the resonance condition when the current through it has its maximum value

The current amplitude for a series, LCR circuit is given by

$Z_{min} = R$

$X_L = X_C$

$(V_0)_L = (V_0)_C$

$I_0 = \frac{\mathcal{E}_0}{\sqrt{R^2+(X_L-X_C)^2}}$

$f_r = \frac{1}{2\pi\sqrt{LC}}$ ↳ natural or resonant frequency

$(I_0)_{max} = \frac{\varepsilon_0}{R}$

Purely resistive

$\phi = 0$

for $(I_0)_{max}$,

$(X_L - X_C) = 0$

$X_L = X_C$

$\omega L = \frac{1}{\omega C}$

$\omega_r = \frac{1}{\sqrt{LC}}$

- Clearly the impedance is minimum.
- The circuit is purely resistive.
- The current and voltage are in the same phase and the current in the circuit is maximum.
- This condition of the LCR-circuit is called resonance condition.
- The frequency at which the current amplitude I_0 attains a peak value is called natural or resonant frequency of the LCR-circuit and is denoted by f.

- Resonance occurs in a series LCK-circuit when

$$X_L = X_C$$

- Resonant frequency,

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

- The impedance is minimum and purely resistive.
- The current has a maximum value of (ε_0/R) at resonant condition.
- The power dissipated in the circuit is maximum and is equal to ε_{rms}^2/R.

$\cos\phi$

- The current is in phase with the voltage or the power factor is unity ($\cos\phi = 1$ when $\phi = 0$).
- Series resonance can occur at all values of resistance R.
- The voltage across R is equal to the applied emf.
- The voltages across L and C are equal and have a phase difference of $180°$ and so their resultant is zero.
- The voltages across L and C are very high as compared to the applied voltage. Hence a series LCR-circuit is used to obtain a large magnification of a.c. voltage.
- The series resonant circuit is also called an acceptor circuit. When a number of frequencies are fed to it, it accepts only one frequency f_r and rejects the other frequencies. The current is maximum for this frequency.

Power in an A.C. Circuit

The rate at which electric energy is consumed in an electric circuit is called its power

$$P = EI$$

$$= (E_0 \sin\omega t)(I_0 \sin(\omega t - \phi))$$

$$= \frac{E_0 I_0}{2}\left[2 \sin\omega t \cdot \sin(\omega t - \phi)\right]$$

$$= \frac{E_0}{\sqrt{2}} \cdot \frac{I_0}{\sqrt{2}}\left[\cos\phi - \cos(\omega t - \phi)\right]$$

$$P_{inst.} = E_{rms} I_{rms}\left[\cos\phi - \cos(\omega t - \phi)\right]$$

for avg. power

$$P_{avg} = E_{rms} I_{rms} \cos\phi$$

$$P_{avg} = P_{apparent} \times \cos\phi$$

↓ Power factor

$$\cos\phi = \frac{R}{Z}$$

POWER FACTOR

The product $\varepsilon_{rms}, I_{rms}$ does not give the actual power and is called apparent power. It gives actual or true power only when multiplied by factor $\cos\phi$. The factor $\cos\phi$ is called the power factor of an a.c. circuit.

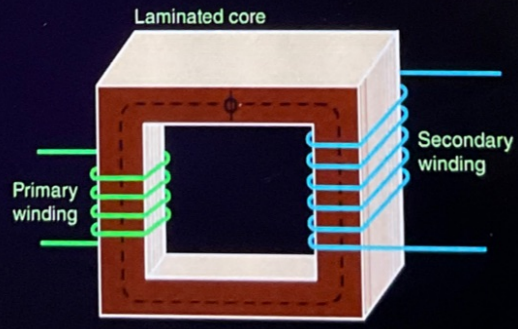
True power = Apparent power × power factor

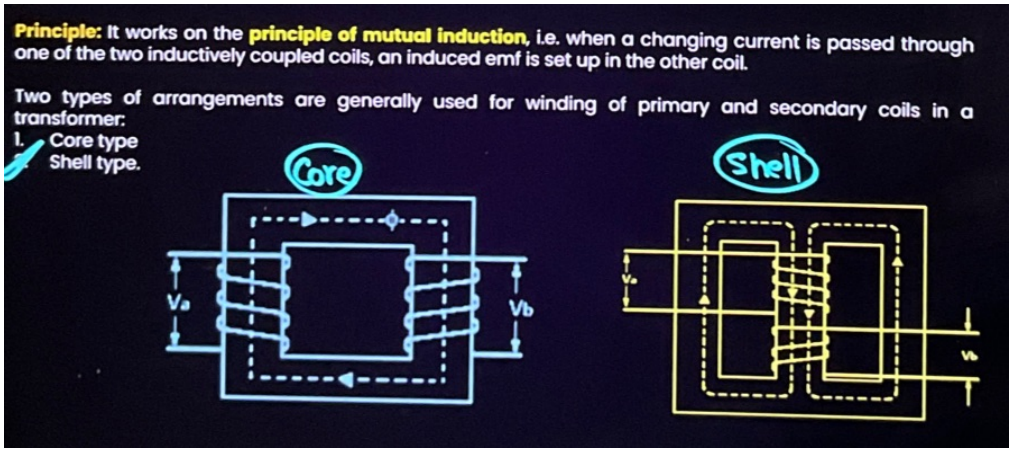
Thus power factor may be defined as the ratio of the true power to the apparent power of an a.c. circuit. **Its value varies from 0 to 1**. The power factor of a series LCR-circuit is given by

$$\cos\phi = \frac{R}{Z} = \frac{R}{\sqrt{R^2 + (\omega L - 1/\omega C)^2}}$$

Transformer

A transformer is an electrical device for converting an alternating current at low voltage into that at high voltage or vice versa. If it increases the input voltage, it is called step up transformer and if it decreases the input voltage, it is called step down transformer.





Consider the situation when no load is connected to the secondary that is its terminals are open. Let N1 and N2 be the number of turns in the primary and secondary respectively.

Then induced EMF in the primary coil, $\epsilon_1 = -N_1 \frac{d\phi}{dt}$

Induced EMF in the secondary coil, $\epsilon_2 = -N_2 \frac{d\phi}{dt}$

Where ϕ is the magnetic flux linked with each turn of the primary or secondary at any instant thus

$$\frac{\epsilon_2}{\epsilon_1} = \frac{N_2}{N_1}$$

The ratio N_2/N_1 of the number of turns in the secondary to that in the primary is called the turns ratio of the transformer. It is also called transformation ratio.

In a step up transformer $N_2 > N_1$ that is the turns ratio is greater than one and therefore $\epsilon_2 > \epsilon_1$ the output voltage is greater than the input voltage

In a step down transformer. $N_2 < N_1$ that is the turns ratio is less than one and therefore. $\epsilon_2 < \epsilon_1$ the output voltage is less than the input voltage.

Currents in Primary & Secondary

assuming the transformer to be ideal and so that there is no energy loses, then

input power = output power

$$\mathcal{E}_1 I_1 = \mathcal{E}_2 I_2$$

where I_1 & I_2 are currents in primary & secondary.

$$\frac{I_1}{I_2} = \frac{\mathcal{E}_2}{\mathcal{E}_1} = \frac{N_2}{N_1}$$

Thus, a step up transformer steps up the voltage but step down the current exactly in the same ratio. Similarly, a step down transformer steps down the voltage but steps up the current exactly in the same ratio.

The efficiency of a transformer is defined as

$$\eta = \frac{\text{Power output}}{\text{Power input}} \times 100\%$$

The efficiency of real transformers is fairly high (90-99%) though not 100%

Energy losses in transformers

The main causes for energy loss in transformers are as follows :

1. **Copper loss:** Some energy is lost due to heating of copper wires used in the primary and secondary windings. This power loss ($= I^2R$) can be minimised by using thick copper wires of low resistance.
2. **Eddy current loss:** The alternating magnetic flux induces eddy currents in the iron core which leads to some energy loss in the form of heat. This loss can be reduced by using laminated iron core.
3. **Hysteresis loss:** The alternating current carries the iron core through cycles of magnetisation and demagnetisation. Work is done in each of these cycles and is lost as heat. This is called hysteresis loss and can be minimised by using core material having narrow hysteresis loop.

4. **Flux leakage:** The magnetic flux produced by the primary may not fully pass through the secondary. Some of the flux may leak into air. This loss can be minimised by winding the primary and secondary coils over one another.

5. **Humming loss:** As the transformer works, its core lengthens and shortens during each cycle of the alternating voltage due to a phenomenon called magnetostriction. This gives rise to a humming sound. So some of the electrical energy is lost in the form of humming sound.

$$\eta = \frac{P_{out}}{P_{in}} \times 100 = \frac{\varepsilon_s I_s}{\varepsilon_p I_p} \times 100$$

Transformer

- A step-up transformer changes a low-voltage into a high-voltage. **This does not violate the law of conservation of energy.** The current decreases by the same proportion. When voltage increases n times, the current reduces to $1/n$ times.
- **A transformer is essentially an a.c. device.** It cannot work on d.c. It changes alternating voltages/currents. It does not affect the frequency of a.c.
- The **small transformers are self-cooled**, which transfer heat directly to the surroundings. **Large transformers are cooled by placing them in oil tanks** to prevent overheating.

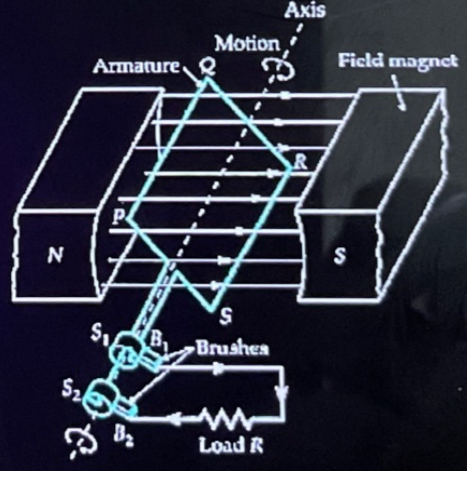
Uses of Transformer

1. Small transformers are used in radio receivers, telephones, loud speakers, etc.
2. In voltage regulators for TV, refrigerators, air-conditioners, computers, etc.
3. In stabilised power supplies.
4. A step-down transformer is used for obtaining large current for electric welding.
5. A step-down transformer is used in induction furnace for melting metals.
6. A step-up transformer is used for the production of X-rays.
7. In the transmission of electric energy from the generating stations to the consumers.

A.C. Generator

A generator or dynamo is a device which converts mechanical energy into electrical energy. Actually, the name generator is a misnomer because it does not generate any energy. It just converts mechanical energy into electrical energy. The present form of a.c. generator is due to Nikoln Tesla who built it in the year 1888.

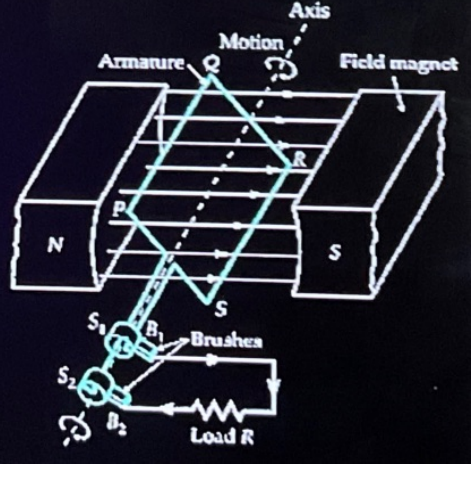
Principle The working of an a.c. generator is based on the principle of electromagnetic induction. When a closed coil is rotated in a uniform magnetic field with its axis perpendicular to the magnetic field, the magnetic flux linked with the coil changes and an induced emf and hence a current is set up in it.



Construction: It essentially consists of the following main parts:

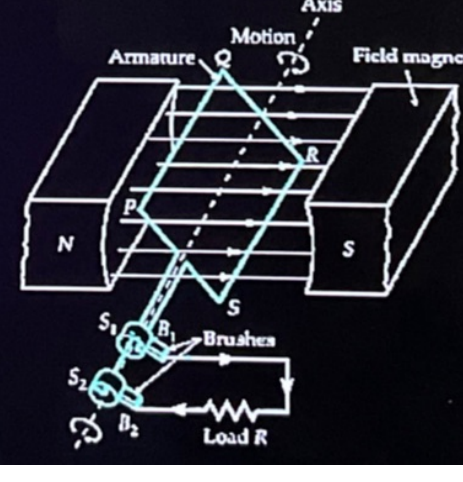
- **Field magnet** It is a permanent magnet of the horse shoe shape in a small dynamo or it is a powerful electromagnet in a large dynamo. It produces a strong magnetic field in the region between its pole-pieces.

- **Armature** It consists of a rectangular coil $PQRS$ having a large number of turns of insulated copper wire wound on a soft iron cylindrical core. The core is laminated one to avoid losses due to eddy currents. The soft iron core concentrates the lines of force to increase the flux density B. The armature can be rotated in the magnetic field of the field magnet about an axis perpendicular to field B.



- **Slip rings:** The two ends of the armature coil are connected to two coaxial brass rings S_1 and S_2 called slip rings. The rings are rigidly fixed to same shaft which is used to rotate the coil. The slips are insulated from each other as well as from the shaft. As the armature coil rotates, the slip rings also rotate about the same axis of rotation.

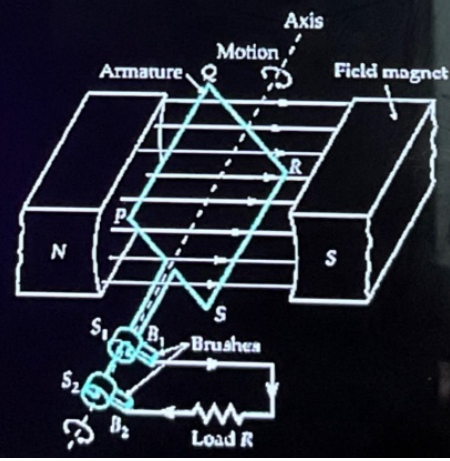
- **Brushes:** Two graphite or flexible metallic rods called brushes are lightly pressed against the two slip rings. The brushes B_1 and B_2 remain fixed in their positions and maintain sliding contacts with the rotatable slip rings S_1 and S_2 respectively. It is through these brushes that the current induced in the armature coil is fed to the external circuit by means of line wires.



Working

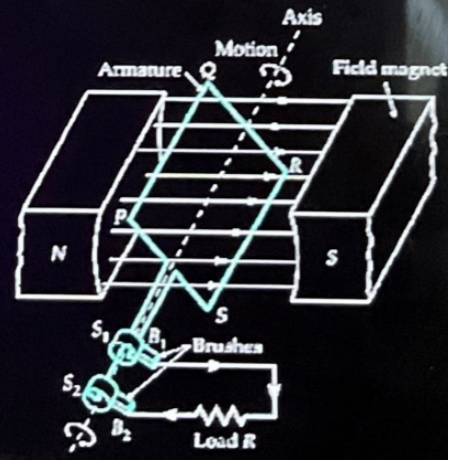
Working

- As the armature coil rotates, the magnetic flux linked with it changes and so an induced current flows through it.
- Suppose initially the coil $PQRS$ be in the vertical position and it is rotated in the **clockwise direction**. The side PQ moves downward and SR moves upward. According to Fleming's right hand rule, the induced current flows from Q to P and from S to R.



Working

- So during the first half rotation of the coil, the induced current flows in the direction **SRQP**, with brush B_1 acting as positive terminal and brush B_2 as negative terminal.
- During the second half-rotation, the side PQ moves upward and SR moves downward. The direction of induced current is reversed, i.e., it flows along **PQRS**, so that the brush B_2 now functions as the positive terminal and brush B_1 as the negative terminal.
- Thus the direction of current in the external circuit is reversed after every half cycle. Hence alternating current is produced by the generator. Such a generator which generates alternating current is called an **a.c. generator or an alternator**.



Expression for Induced E.M.F

$\phi_B = BA\cos\omega t$

By EMI,

$$\mathcal{E}_{in} = -\frac{d\phi_B}{dt}$$

$$= -\frac{d}{dt}(BA\cos\omega t)$$

$$= -BA(-\sin\omega t \cdot \omega)$$

$$\mathcal{E}_{in} = +BA\omega\sin\omega t$$

for N turns,

$$\mathcal{E}_{in} = NBA\omega\sin\omega t = \mathcal{E}_0\sin\omega t.$$

$$I = \frac{\mathcal{E}}{R} = \frac{\mathcal{E}_0\sin\omega t}{R}$$

$$I = I_0\sin\omega t.$$

Advantages & disadvantages of A.C. over D.C.

Advantages

1. The generation of AC is more economical than DC
2. The alternating voltage can be easily stepped up or step down by using a transformer
3. The alternating currents can be reduced by using a choke coil without any significant wastage of energy
4. The alternating currents can be transmitted to distant places without any significant line loss
5. Also, AC can be easily converted into the DC by using rectifiers
6. AC machines are simple and robust and do not require much attention during their use

Disadvantages

1. Peak value of a AC is high ($I_0 = \sqrt{2} I_{rms}$) It is dangerous to work with AC
2. In phenomena like electroplating, electro refining, electro typing, AC cannot be used
3. AC is transported more from the surface of a conductor than from inside. This is called skin effect. Therefore several fine, insulated wires, (and not a single thick wire) are required for the transmission of AC

THANK YOU !!